

Gerardo Sanchez

Modern Engineer

Full stack in the modern sense: data, cloud, web and AI,
from the warehouse to the screen.

Experience

Independent Data & AI Engineer · Corporate clients OCT 2025 - PRESENT

- Designed a serverless platform on Google Cloud (Cloud Functions, Pub/Sub, Cloud Storage, BigQuery, Secret Manager, budget controls) that brings sales, marketing and operations data into one warehouse.
- Built a Python webhook service that sends offline CRM sales to the Meta Conversions API, tying revenue to ad spend for a business with no pixel and no prior attribution. In production within a week.
- Modelled the warehouse in dbt with quality tests and delivered it as executive dashboards in Looker Studio and Power BI.
- Traced a funnel from ad to website, message, appointment and sale to find where volume was lost, and ranked the fixes by impact.
- Own and operate everything I ship on a monthly service agreement; every project so far started as an analysis I proposed to management.

Lead Data Analyst · TortiRegias · Nyjisa · Chilokia JAN 2024 - OCT 2025

- Owned the group-wide Alpha ERP rollout with every department head: process documentation, data traceability and quality controls, and the SQL (MS SQL Server, MySQL) reconciling purchasing, sales, production and accounting.
- Built App Tarimas, a React drag-and-drop app that generates Walmart CEDIS pallet labels with scannable barcodes. An impractically slow manual task now takes about two minutes.
- Built the Power BI production dashboard and Dashboard de Mantenimiento, plus Streamlit views, giving management its first live look at operations; presented findings weekly.
- Designed a manufacturing cost model inside the ERP that surfaced unit cost and margin per product for the first time.
- Automated recurring PDF reports and payroll receipts in Python, removing about two months of manual work a year.

Undergraduate Researcher · UANL, Facultad de Ciencias Físico Matemáticas 2022 - 2025

- Thesis on gravitational microlensing: derived the lensing model from first principles, simulated it in Fortran across a wide parameter sweep, and produced 223k labelled light curves to train a classifier that detects microlensing events.

Education

B.S. Physics

UANL, Monterrey · 2019 — 2023

AI & Machine

Learning
Caltech CTME · 2024

Applied Data Science

Simplilearn · 2024

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github.com/PlayingArc

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Monterrey, Mexico

Open to remote & relocation

Skills

Languages

Python, TypeScript, JavaScript, SQL, Bash, Fortran

Data & BI

BigQuery, dbt, PostgreSQL, MySQL, MS SQL Server, Power BI, Looker Studio, Streamlit, pandas

Cloud & Web

Google Cloud (Cloud Run, Functions, Pub/Sub, Secret Manager), Cloudflare, Next.js, React, REST, OAuth, webhooks

AI

OpenAI & Claude APIs, LLM classification with structured output, multi-agent tooling, scikit-learn

Practice

Git, GitHub Actions, Linux, ADRs, AI-assisted development

Selected Projects

Money on Rails

Budget analysis on the YNAB API, built and run alone on Cloud Run and serverless Postgres, with an LLM layer that classifies categories.

App Tarimas

Pallet-label generator for Walmart CEDIS, from hours of manual work to two minutes.

Personal agent system

Self-hosted multi-agent assistant with MCP tools, scoped Google access and approval gates on anything outward-facing.

Languages

Spanish, native
English, TOEFL iBT 98
Japanese, JLPT N5